

## Improving mixing efficiency in laminar-flow static mixers with baffle inserts and vortex generators: A three-dimensional numerical investigation using corrugated tubes

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### ABSTRACT

Static mixers are vital for continuous processes, improving product quality, reducing costs, and enhancing efficiency. This study numerically investigates the mixing performance of various geometry configurations in static mixers. A corrugated tube with baffle inserts arranged in a triangular pattern is used. These studies are carried out using the CFD COMSOL Multiphysics 6.1 software with Reynolds numbers from 10 to 300, examining the impact of vortex generators on hydrodynamic behavior. The obtained outcomes reveal that the implementation of a corrugated tube induces a proficient recirculation flow within the divergence zone, thereby augmenting mixing efficiency by three times in comparison to a **straight** pipe as indicated by the mixing index. Introducing a baffle configuration composed of four triangular elements yields a heightened mixing index, albeit at the cost of an elevated pressure drop. An alteration of the angle of attack to 30° leads to a notable 22.6% enhancement in the mixing index while concurrently reducing pressure drop, in contrast to the condition where  $\beta$  equals 0°. These findings hold valuable implications for the optimization of static mixer designs, thereby elevating their utility within laminar flow applications. The resultant benefits encompass improved product quality, diminished operational costs, and heightened process efficiency.

### 1. Introduction

Achieving perfect mixing is a critical requirement for many industries to ensure the quality of their finished products. These industries range from agricultural, pharmaceutical, and polymer to chemical and petrochemical, and other technical applications. Despite the widespread need for perfect mixing, many processes still rely on batch mixing with stirred vessels as the primary mixing model, which has significant drawbacks. These include locally high shear rates, large space requirements, high maintenance costs, and long residence time, which can negatively impact sensitive materials [1–3]. As a result, there is a need for more efficient and effective mixing techniques that can improve product quality and reduce costs while addressing these challenges.

Static mixers have emerged as a favored substitute for dynamic mixing within continuous processes owing to a multitude of advantages.

Principally, the absence of mobile components in static mixers engenders heightened reliability and reduced susceptibility to mechanical deterioration. Moreover, static mixers necessitate a smaller spatial footprint and entail diminished capital outlays when contrasted with their dynamic counterparts. The self-sufficiency of static mixers, wherein the requisite power is supplied by the inline flow's propulsive pump, obviates the necessity for an auxiliary motor drive, thereby imparting an additional layer of cost reduction. In static mixers, the mixing process is achieved by continuously dividing, extending, and transposing fluids using motionless elements. Static mixers have evolved to become a critical tool in process engineering and are widely applied in both turbulent and laminar flow regimes to enhance mass and heat transfer, including blending, reaction, dispersion, and emulsion [4,5]. As a result, static mixers are an attractive option for many industries looking to improve their mixing processes and reduce costs.

Passive methods have been developed and applied in various

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| Nomenclature         |                                                  | x,y,z                      | [m] coordinates                                           |
|----------------------|--------------------------------------------------|----------------------------|-----------------------------------------------------------|
|                      |                                                  | f                          | [-] friction factor                                       |
| <i>Greek symbols</i> |                                                  |                            |                                                           |
| $\bar{c}$            | [mol/m <sup>3</sup> ] average concentration      | $\beta$                    | [°] angle of attack                                       |
| $\bar{v}$            | [m/s] average velocity of fluid                  | $\mu$                      | [kg/m s] dynamic viscosity                                |
| $\Delta p$           | [Pa] pressure drop in a static mixer             | $\rho$                     | [kg/m <sup>3</sup> ] density of fluid                     |
| A                    | [mm] amplitude of wavy channel                   | $\sigma$                   | [mol/m <sup>3</sup> ] standard deviation of concentration |
| B                    | [mm] distance between two elements               | <i>Abbreviated symbols</i> |                                                           |
| C                    | [mol/m <sup>3</sup> ] concentration              | MI                         | mixing index, dimensionless                               |
| $c_i$                | [mol/m <sup>3</sup> ] concentration of species   | CoV                        | coefficient of variance, dimensionless                    |
| D                    | [mm] diameter of tube                            | CFD                        | computational fluid dynamics                              |
| $d_h$                | [mm] diameter hydraulic                          | TS                         | triangles staggered                                       |
| $D_i$                | [mol/m <sup>3</sup> ] diffusion coefficient      | TA                         | triangles aligned                                         |
| G                    | [m/s <sup>2</sup> ] gravity acceleration         | VGs                        | vortex generators                                         |
| N                    | [-] total number of samples in the cross-section |                            |                                                           |
| P                    | [Pa] static pressure                             |                            |                                                           |
| Re                   | [-] Reynolds number                              |                            |                                                           |
| u                    | [m/s] axial velocity                             |                            |                                                           |

industrial fields to enhance mass and heat transfer in continuous processes. One such method is the use of VGs, which have gained widespread application. VGs come in several types, including helical inserts, curved channels, flexible and rigid baffles [6]. The primary objective of VGs is to agitate the flow field by creating a radial flow between the center flow and the near-wall fluids [7]. This radial flow enhances the mixing performance by increasing turbulence and promoting the formation of eddies, which improve the mass and heat transfer. The use of VGs offers a cost-effective solution to enhance mixing performance while minimizing energy consumption and maintaining product quality.

Numerous studies have investigated the impact of VGs on heat and mass transfer efficiency, utilizing both experimental and numerical methods. For example, Mokrani et al. [8] developed a trapezoidal baffle with 90° tabs that were angled 30° relative to the wall. By measuring the flow with LDV, they demonstrated that longitudinal vortices have a significant impact on mixing. These vortices were found to increase turbulence intensity from 5% to 17%. Hosseini et al. [9] conducted a separate investigation into the efficiency of static mixers featuring 6, 8, and 10 mixing elements, each comprising four blades that were rotated 45° from the previous one. Their results showed that the use of 10 mixing elements resulted in a coefficient of variation of less than 0.05, indicating that the mixing performance was strong.

Vikhansky et al. [10] conducted research on a recently developed high efficiency vortex (HEV) static mixer to evaluate its emulsification effectiveness. They placed baffles on the walls of the mixer to generate large longitudinal vortices that improved transversal mixing and facilitated droplet breaking along the wall. Another investigation conducted by Kaid et al. [11] analyzed the efficiency of a newly designed static mixer consisting of a convergent/divergent circular tube equipped with staggered baffles. The researchers employed CFD tools to examine various parameters such as mixing quality and pressure drop. The study indicated that the diverging technique reduced pressure while maintaining a high mixing index. Additionally, Mahmoodi et al. [12] altered the design of a flat-blade static mixer by introducing a novel curved-blade mixer with 8, 10, and 14 mixing elements at *Re* ranging from 600 to 7000. The findings revealed a decline in mixer pressure drop of up to 23% and an enhancement in mixing quality resulting from the modification.

Habchi et al. [13] carried out a numerical study on chaotic mixing by utilizing multiple VG arrays positioned inside a circular pipe. Their simulations focused on two distinct configurations: a first configuration in which the VGs were aligned, and a second configuration in which they were rotated 90° relative to the streamwise axis. The study revealed that VGs, particularly in the alternative configuration, were highly effective in improving mixing performance. Zhou et al. [14] conducted an

investigation on a novel static mixer, which aimed to improve turbulent reactive mixing within a tubular turbulent flow. Hamidouche et al. [15] conducted experimental research using stereoscopic particle image velocimetry (PIV) to investigate the flow behavior downstream of delta-winglet VGs in a range of *Re* from 400 to 12,000. Their findings suggest that the mixing process is more effective at low *Re* than at high *Re*. In a separate study, Zheng et al. [16] examined the effects of turbulence on particles of different sizes in a tube with and without VGs at varying *Re*. They identified a correlation between turbulence and particle size, called the flotation rate constant, and determined that the highest separation inlet velocity occurs at a flotation rate of 0.3. Aravind et al. [17] numerically assessed the enhancement of convective mass transfer in compressible flow using lateral sweep angles of a longitudinal swirl generator. The results showed that the vorticity increases as the sweep angle increases, leading to an improvement in mass transfer. These studies highlight the importance of understanding the impact of VGs on different types of fluids and particles, as well as the *Re* and geometrical parameters of the VGs, in optimizing the mixing and mass transfer efficiency.

Boonloi et al. [18] used numerical analysis to examine the performance of double-V baffles in a circular tube with an isothermal wall. The researchers evaluated the effectiveness of the baffle using two metrics, the Nusselt number (Nu) and the thermal improvement (TEF), and tested different combinations of double-V baffles to find the optimal configuration. The findings showed that the optimal values of Nu and TEF were 3.55 and 22.42, respectively. Meanwhile, Nuthvipa et al. [19] conducted an experimental investigation into the turbulent flow behavior and heat transfer of a circular tube fitted with V-shaped baffles inserted on two sides of a flat sheet. The researchers made modifications to the baffle by adding louvers and controlling its angle. Various researchers have explored the application of flexible VGs in enhancing heat and mass transfer. For instance, Amini et al. [20] conducted a parametric study to examine the impact of flexible splitter-equipped VGs on heat transfer performance. Ali et al. [21] investigated the impact of varying the number and position of flexible VGs on heat transfer and mixing at a fixed *Re* of 2000. The study compared the results of both flexible and rigid configurations and found that the flexible flaps significantly improved mixing by up to 99% compared to the rigid cases.

The application of VGs is not limited to macro-scale systems only but is also widely used in microfluidic systems. In a study conducted by Hsiao et al. [22], the effect of VGs on the mixing of two fluids in a T-shaped microchannel was investigated. They utilized the Taguchi method to determine the optimal combination of geometrical parameters that could enhance the mixing process, and the findings suggested that the angle of attack and winglet height were the most crucial factors.

Another study proposed a planar micromixer with V-shaped baffles by Yuan et al. [23]. They performed both numerical and experimental analyses to evaluate the mixing efficiency across  $Re$  from 0.1 to 50. The results indicated that the position and direction of the baffle played a significant role in generating vortices and enhancing the mixing performance. These studies demonstrate the potential of VGs in microfluidic systems and the importance of optimizing their geometrical parameters to improve mixing efficiency.

A distinctive facet underscoring the innovativeness of this investigation is its concentration on the utilization of corrugated tubes outfitted with triangular baffles to enhance mixing efficacy. While the scholarly literature has extensively scrutinized baffles, the employment of corrugated tubes as a mechanism for amplifying mixing within meso-scale systems remains largely unexplored. By delving into the effects of varying parameters such as  $Re$ , wave amplitude, position, the number of baffles, and their angle of attack on the mixing behavior of Newtonian flows within corrugated tubes, this study endeavors to furnish novel insights into the realm of mixing enhancement. Furthermore, this research introduces an additional distinctive facet through the utilization of numerical modeling to simulate flow dynamics and mixing phenomena, thereby offering a more intricate comprehension of these fluidic processes. The ultimate aspiration of this research is to yield consequential practical implications, particularly for industries encompassing chemical and biochemical engineering, where optimal mixing

profoundly influences the performance and yield of diverse processes.

The organization of this study is clear and concise, with Section 2 detailing the numerical modeling approach, Section 3 investigating the impact of different parameters on mixing efficiency, and Section 4 summarizing the conclusions. The findings of this research can offer valuable insights into meso-mixing and its potential use in continuous processes.

## 2. Modeling and simulation

### 2.1. Problem presentation

Fig. 1 illustrates the proposed design of the static mixer with horizontal sinusoidal-wave channels, which is equipped with staggered and aligned baffle inserts.

The mixer comprises six mixing elements that are placed on the wave crests of the channel. Each element contains triangle-shaped baffles, which are rotated at angles of  $45^\circ$  and  $90^\circ$  with respect to the neighboring elements, resulting in staggered arrangements of 4-Triangles and 2-Triangles, respectively. The static mixer's key geometric parameters comprise the length of the tube ( $L$ ), its diameter ( $D$ ), the distance ( $b$ ) between the baffles (representing the mixing unit), and the crest diameter of the tube ( $D_c$ ). Table 1 lists the geometric parameters of the static mixer. The x coordinate represents the primary flow direction, while the

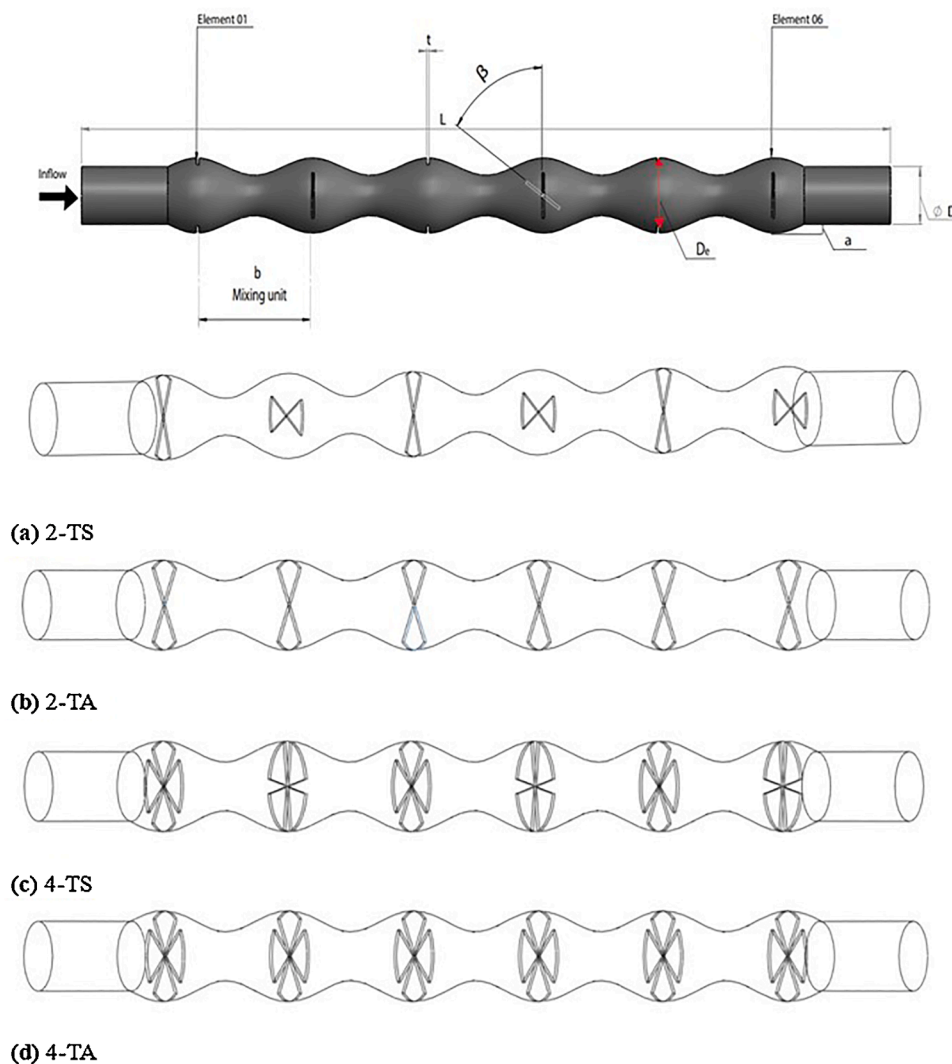


Fig. 1. Schematics of wave tubes with staggered (TS) and aligned (TA) triangular-baffles.

**Table 1**

The geometrical dimensions of the static mixer.

| Quantity              | Symbol         | Value      |
|-----------------------|----------------|------------|
| Pipe diameter         | D              | 20 mm      |
| Length of the channel | L              | 350 mm     |
| Wavy channel diameter | D <sub>e</sub> | 26 mm      |
| Wave amplitude        | a              | 0 and 3 mm |
| Step of element       | b              | 50 mm      |
| Angle of attack       | β              | 0° - 45°   |
| Thickness of baffle   | t              | 1mm        |

transversal coordinates are represented by the y and z coordinates.

The static mixer design with corrugated tubes and triangle baffles presented in this study is unique and has not received much attention in the literature. The objective of this research is to conduct a numerical analysis to assess the mixer's performance and to study the impact of different parameters, such as *Re*, wave amplitude, baffle position, and number and angle of attack.

## 2.2. Boundary conditions

The present research uses computation fluid dynamics (CFD) to analyses a three-dimensional laminar flow and the dilute mass transport module. The following operating conditions were utilized in the simulations:

- The simulations are three-dimensional and steady.
- The fluid-flow regime is laminar.
- The fluid is incompressible and Newtonian.
- The velocity at the inlet of the channel is assumed to be a constant and uniform value, denoted as  $U_{\text{int}}$ .
- The liquid concentrations in the two inlets of the tube were fixed at  $c_0 = 0$  and  $c_1 = 10 \text{ mol/m}^3$ .
- The boundary at the outlet is specified as the pressure outlet, where the pressure is set to 0 Pa.
- The wall was set with the no-slip condition as a stationary wall.

In this study, the mixing behavior of fluids was simulated using pure water and dye-water. Water without any impurities (pure) was selected as the working fluid because of its well-established physical properties such as a density of  $997 \text{ kg/m}^3$  and a dynamic viscosity of  $0.001 \text{ kg/m.s}$ . To track the mixing process, dye-water was used, which has a diffusion coefficient of  $10^{-9} \text{ m}^2 \cdot \text{s}^{-1}$  [10]. The simulations were conducted under laminar flow conditions, and the input velocity was adjusted by modifying the *Re*. The VG, represented by the baffle inserts, was found to be the most critical factor influencing mixing performance.

## 2.3. Governing equations

The static mixer operates on the assumption that the fluids involved are Newtonian fluids. This means that they exhibit a constant viscosity regardless of the shear rate applied. Additionally, the fluids are considered to be steady, incompressible and laminar, meaning that the body force is negligible. The equations that govern the system under these conditions consist of the continuity equation, the Navier-Stokes equation, and the Convection-Diffusion equation, which are expressed as follows [24]:

$$\nabla \cdot \vec{V} = 0 \quad (1)$$

$$\rho \vec{V} \cdot \nabla \vec{V} = -\nabla P + \mu \nabla^2 \vec{V} \quad (2)$$

$$\vec{V} \cdot \nabla C_i = D_i \nabla^2 C_i \quad (3)$$

The fluid behavior in the static mixer is described by a set of equations that includes parameters such as fluid velocity (*V*), fluid density

( $\rho$ ), dynamic viscosity ( $\mu$ ), static pressure (*P*), species concentration ( $C_i$ ), and mass diffusion coefficient ( $D_i$ ) for each species 'i' in the mixing.

The *Re* number is a crucial parameter for the analysis of fluid flow and mixing. It is defined as:

$$Re = \frac{\rho V D_h}{\mu} \quad (4)$$

The friction factor defined as:

$$f = \frac{\Delta P}{\frac{1}{2} \rho V^2} \frac{D_h}{L} \quad (5)$$

The term  $D_h$ , when referring to a cylinder, denotes the pipe diameter. The pressure difference between the inlet and outlet is denoted as  $\Delta P$ , whereas *L* refers to the length of the mixer.

## 2.4. Mixing index

Mixing homogeneity can be evaluated using various parameters such as temperature [25] and concentration [26] profile. However, assessing homogeneity based on an intuitive sense can be challenging. Therefore, a quantitative method is used, where the mixing quality is typically evaluated in terms of a mixing index. This index is determined by conducting a statistical analysis of mixture samples at the cross-section of the channel's outflow using the following equations [11,27,28]:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (C_i - \bar{C})^2} \quad (6)$$

$$MI = 1 - \frac{\sigma}{\bar{C}} = 1 - \text{COV} \quad (7)$$

The standard deviation of concentration at a specific cross-section is denoted by  $\sigma$ .  $C_i$  refers to the concentration of the analyzed species, *N* is the total number of samples, and  $\bar{C}$  is the mean concentration, which is computed using the following formula:

$$\bar{C} = \frac{1}{N} \sum_{i=1}^N C_i \quad (8)$$

The mixing index is a valuable parameter for assessing the homogeneity of a mixture. When the mixing index value is 0, it indicates that the fluids are completely unmixed, while a value of 1 ( $\sigma = 0$ ) means that the two species are entirely mixed. In mixing-process applications, a mixing efficiency of above 90% is often deemed acceptable.

The extent of mixing homogeneity can be assessed using the coefficient of variation (Cov) which is the ratio of the standard deviation of concentration ( $\sigma$ ) to the average concentration ( $\bar{C}$ ) [26]. The Cov can be mathematically represented as:

$$COV = \frac{\sigma}{\bar{C}} = \frac{1}{\bar{C}} \sqrt{\frac{1}{N} \sum_{i=1}^N (C_i - \bar{C})^2} \quad (9)$$

When COV approaches 0, it indicates that the fluid is well mixed, and when COV approaches 1, it indicates that the fluid is poorly mixed.

## 2.5. Achieving grid independence

In order to assess the grid independence of CFD simulations, an unstructured mesh was generated using tetrahedral elements in CFD software. Four different meshes were created to simulate the flow of a Newtonian fluid under constant laminar flow conditions at *Re* = 100. The number of elements varied across the four tests, with mesh1 containing 347,822 elements and mesh4 containing 4,639,490 elements. Table 2 summarizes the results of the mesh tests conducted to determine the optimal grid size for  $\Delta P$  and MI values. The difference in these values between mesh3 and the final mesh was found to be minimal, at only 1%



**Table 2**

Assessment of mesh independence.

|                    | Mesh1   | Mesh2   | Mesh3     | Mesh4     |
|--------------------|---------|---------|-----------|-----------|
| Element number     | 347,822 | 620,495 | 1,532,209 | 4,639,490 |
| Pressure drop [Pa] | 0.8024  | 0.7964  | 0.7853    | 0.7771    |
| MI                 | 0.9645  | 0.9768  | 0.9871    | 0.9877    |
| CPU time (second)  | 10,520  | 14,881  | 86,847    | 269,225.7 |

and 0.06%, respectively.

Furthermore, Fig. 2(a) and (b) illustrate the variations in velocity magnitude and concentration across the channel, respectively, demonstrating no significant difference between mesh3 and mesh4. Based on these findings, it can be concluded that mesh3 provided the best grid size for the calculation.

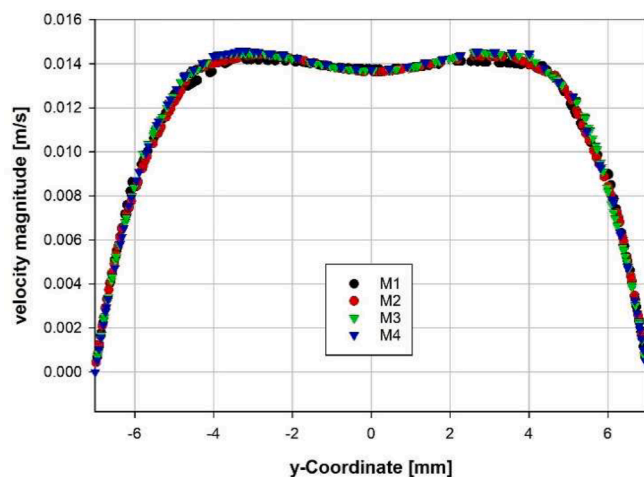
## 2.6. Numerical diffusion

The choice of an effective numerical discretization strategy is crucial, contingent upon the verification of precision, dependability, and consistency of numerical results achieved by augmenting mesh density. The primary objective is to mitigate the most prevalent sources of numerical errors in CFD, particularly numerical diffusion. The accuracy and behavior of the numerical solution are directly correlated with the selection of shape functions in finite element methods. To explore numerical diffusion and determine the optimal discretization method for convective-diffusion terms, two discretization schemes (linear and quadratic basis functions) have been investigated. The outcomes of this comparison are presented in (refer to Fig. 3).

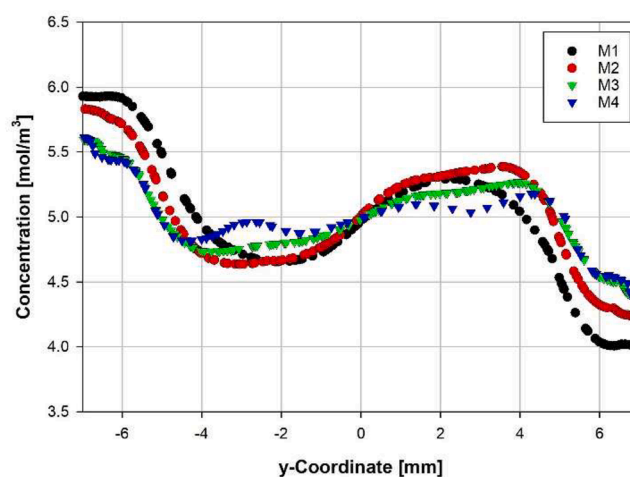
Clearly, the discretization methods employed for species transfer yield divergent outcomes, with the disparities being most pronounced in the initial half of the mixer. Owing to elevated numerical discretization errors, the utilization of linear basis functions in this region exhibits a propensity to overestimate mixing performance. Conversely, the implementation of quadratic basis functions yields results that are both accurate and quantifiable. Altering the discretization forms of basis functions has a minor impact on numerical diffusion under circumstances such as baffled tubes and intricate designs [29].

## 2.7. Numerical validation

In this study, there were no prior research papers or direct experimental data available on the mixing and hydrodynamic characteristics of the corrugated tube with baffles.



(a)



(b)

Fig. 2. Variation of (a) velocity magnitude, (b) concentration, at  $x = 175$  mm (middle) for various grid sizes,  $Re = 100$ .

In order to validate the accuracy of the simulation method used in this study, the CoV coefficient for the mixing of two Newtonian fluids was compared to the experimental studies obtained by Al-Atabi [30]. Al-Atabi studied the mixing of fluids in a circular tube with baffle insets. The comparison was made for two cases of baffles in a static mixer, covering a range of  $Re$  from 100 to 600. The results of this comparison are presented in Fig. 4(a), which shows excellent agreement between the CFD data and the experimental results.

Furthermore, to validate the model, the friction factor of water in the smooth channel was calculated using the Darcy equation (Eq. (5)). As depicted in Fig. 4(b), the numerical findings show reasonable agreement with the theoretical predictions. Thus, the current CFD approach is deemed appropriate for investigating other static mixers with a satisfactory level of accuracy.

One can obtain precise calculations of the friction factor for laminar flows across a smooth tube with a constant wall temperature by utilizing the solutions given below [31]:

$$f = \frac{64}{Re} \quad (10)$$

## 3. Results and discussion

The investigation was initiated by examining the influence of flow rates on the mixing process, which were quantified by the  $Re$  value. The  $Re$  number was limited to the laminar regime, and the values of  $Re$  were gradually increased to generate higher interfacial areas and improve mixing while decreasing the residence time. The  $Re$  range examined in the study was from 10 to 300. The aim of the study was to examine how various flow rates affect the mixing process and to identify the ideal flow conditions that can improve mixing efficiency and decrease residence time.

### 3.1. Effects of flow rates

Fig. 5(a) and (b) depict the axial velocity curves of the flow in a static mixer with 2-TA at varying  $Re$  numbers. It was observed that an increase in the  $Re$  number has a significant impact on the velocity evolution in a static mixer. Interestingly, all curves exhibited the same pattern despite varying values, indicating conformity to the principle of mass continuity. Examination of the axial spatial configuration of the geometry unveiled a notable augmentation in velocity within the convergent region of the channel, attributed to the alteration in diameter, a widely acknowledged phenomenon in fluid mechanics. Conversely, the radial

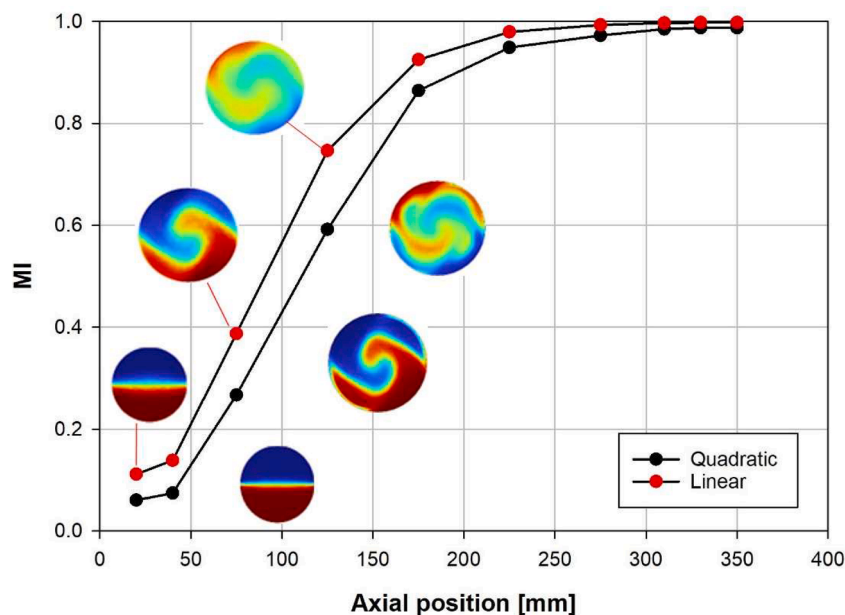
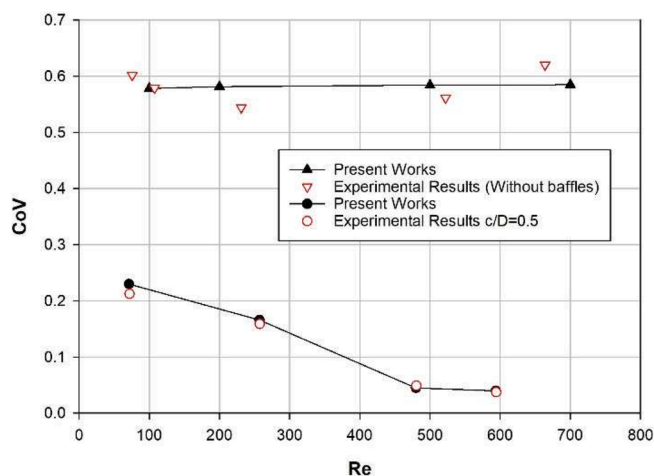
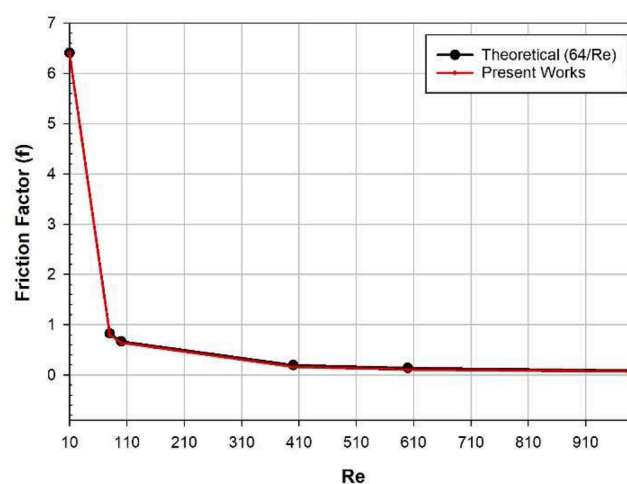


Fig. 3. Comparison of the Mixing index (MI) along mixer for different discretization schemes for the configuration of 4-TA 30°,  $Re = 100$ .



(a)



(b)

Fig. 4. Validation results: (a) Comparison between simulation and experimental values of CoV, (b) Friction factor in a smooth pipe for laminar flow.

placement and morphology of the baffles exert substantial influence upon the distribution of velocity. Remarkably, the velocity profile exhibits a prominent peak at the central region immediately following the baffle, accompanied by an exclusive occurrence of a negative value within the divergence domain of the channel. This observation suggests the existence of a recirculation bubble in this region. The insights gleaned from our analysis of the flow behavior in a static mixer with 2-TA can inform the development of more efficient mixing systems, highlighting the importance of understanding the impact of  $Re$  numbers and geometry on fluid flow behavior.

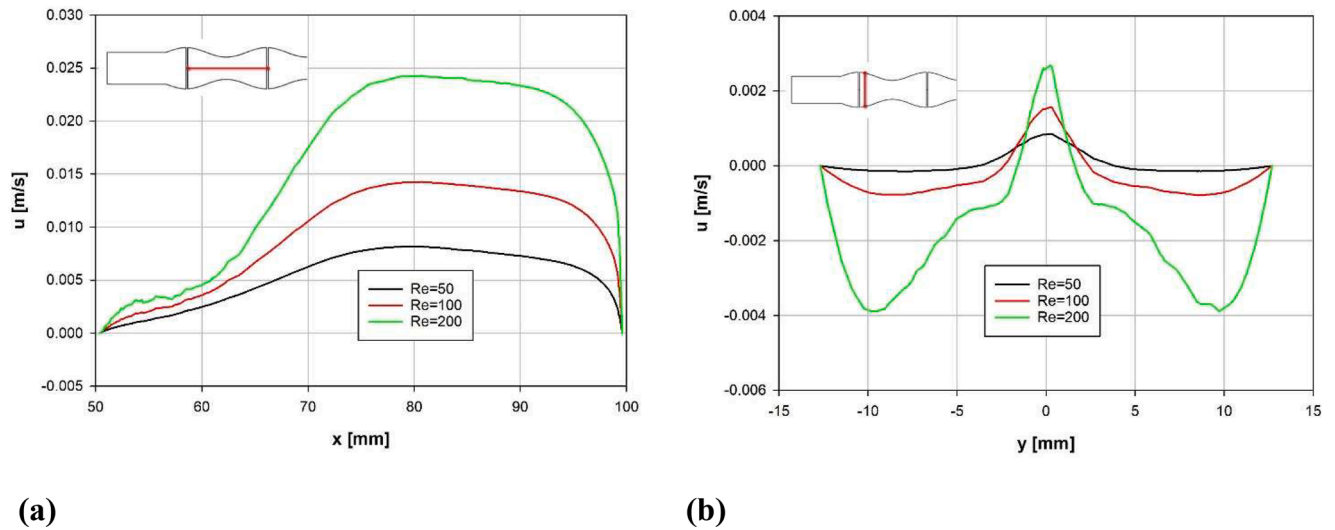
Fig. 6 displays the axial velocity profile between the first and third elements of the static mixer with 2-TA. The plot of the flow streamlines is displayed on the central horizontal plane of the geometry, specifically at  $z/D = 0.5$ . One notable finding from the visualization is that when the  $Re$  number is low (50), the flow adheres to the intended path and creates a low noteworthy primary or secondary vortex. However, as the  $Re$  number increases, vortex formations begin to increase and develop behind and before the baffles. This phenomenon contributes to

improved mixing and enhances the performance of the static mixer.

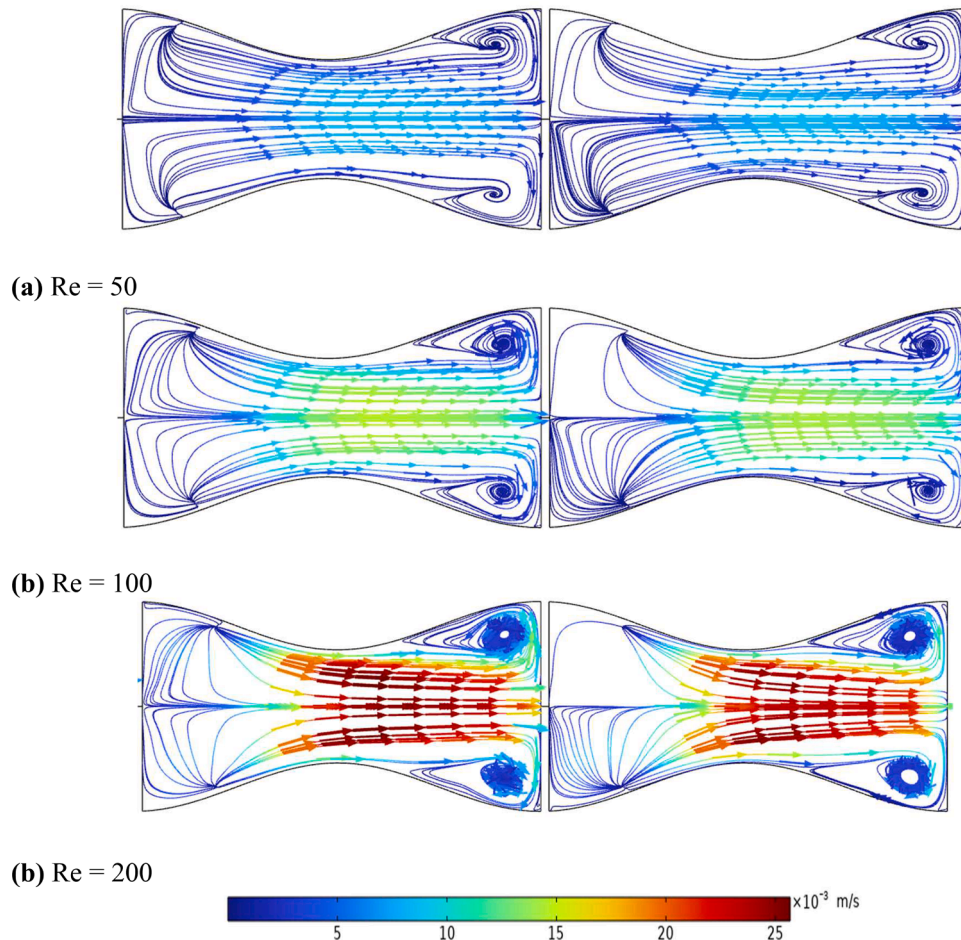
The findings presented in Fig. 6 highlight the importance of  $Re$  number in determining the flow behavior and mixing performance of the static mixer with 2-TA. Such insights can be leveraged to optimize the design of static mixers for various applications, taking into account the impact of  $Re$  number on the flow behavior.

### 3.2. Influence of amplitude

In the context of a static mixer with 2-TA, Fig. 7(a) and (b) display the influence of the amplitude height on the flow's mixing index and pressure drop over a broad range of  $Re$  numbers. The results of the investigation reveal a discernible correlation between mixing efficiency and the  $Re$  number, elucidating an 11.3% enhancement in the mixing index within the corrugated tube configuration in comparison to the straight tube at lower  $Re$  numbers ( $Re < 50$ ). This substantiates the efficacy of corrugated tubes in facilitating mixing by attenuating the efficacy of vortex formation both upstream and downstream of the baffle.



**Fig. 5.** The axial velocity profiles of the flow in a static mixer with 2-TA,  $\beta=0^\circ$  in (a): the axial position between the first and second elements ( $y = 0$  and  $z = 0$ ), (b) the radial position behind the first element ( $x = 53$  mm).



**Fig. 6.** The streamline plot at  $z/D = 0.5$  between elements 1 and 3 of the static mixer with 2-TA,  $\beta = 0^\circ$ .

As the  $Re$  number increases ( $Re > 50$ ), vortex formations in the corrugated channel become more prominent, especially before the baffles. This phenomenon is evident from the improved efficiency of the corrugated channel by a 3-fold mixing index increase compared to a straight tube, where vortices form only behind the baffles. The study's

results underscore the significance of amplitude height in shaping the pressure drop and mixing effectiveness of the flow in the static mixer equipped with 2-TA, spanning various Reynolds numbers.

Corrugated tubes have been widely used to enhance heat and mass transfer, but they also cause an increase in pressure drop as compared to



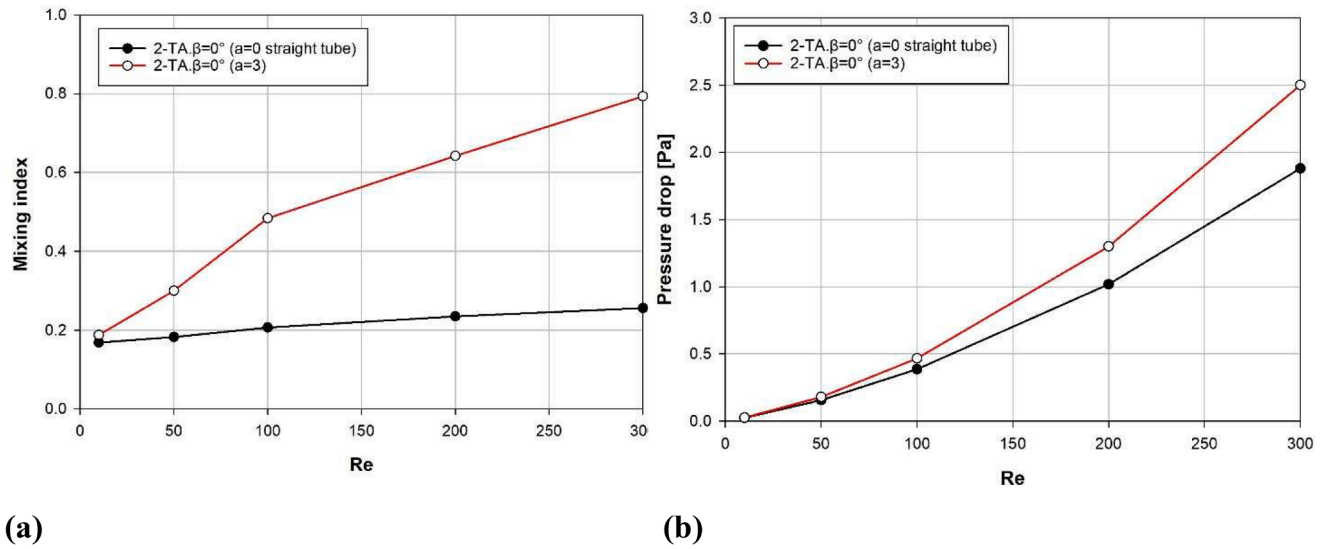


Fig. 7. The impact of amplitude on two critical performance parameters of the static mixer with 2-TA,  $\beta = 0^\circ$ : (a) mixing index at the outlet, and (b) pressure drop.

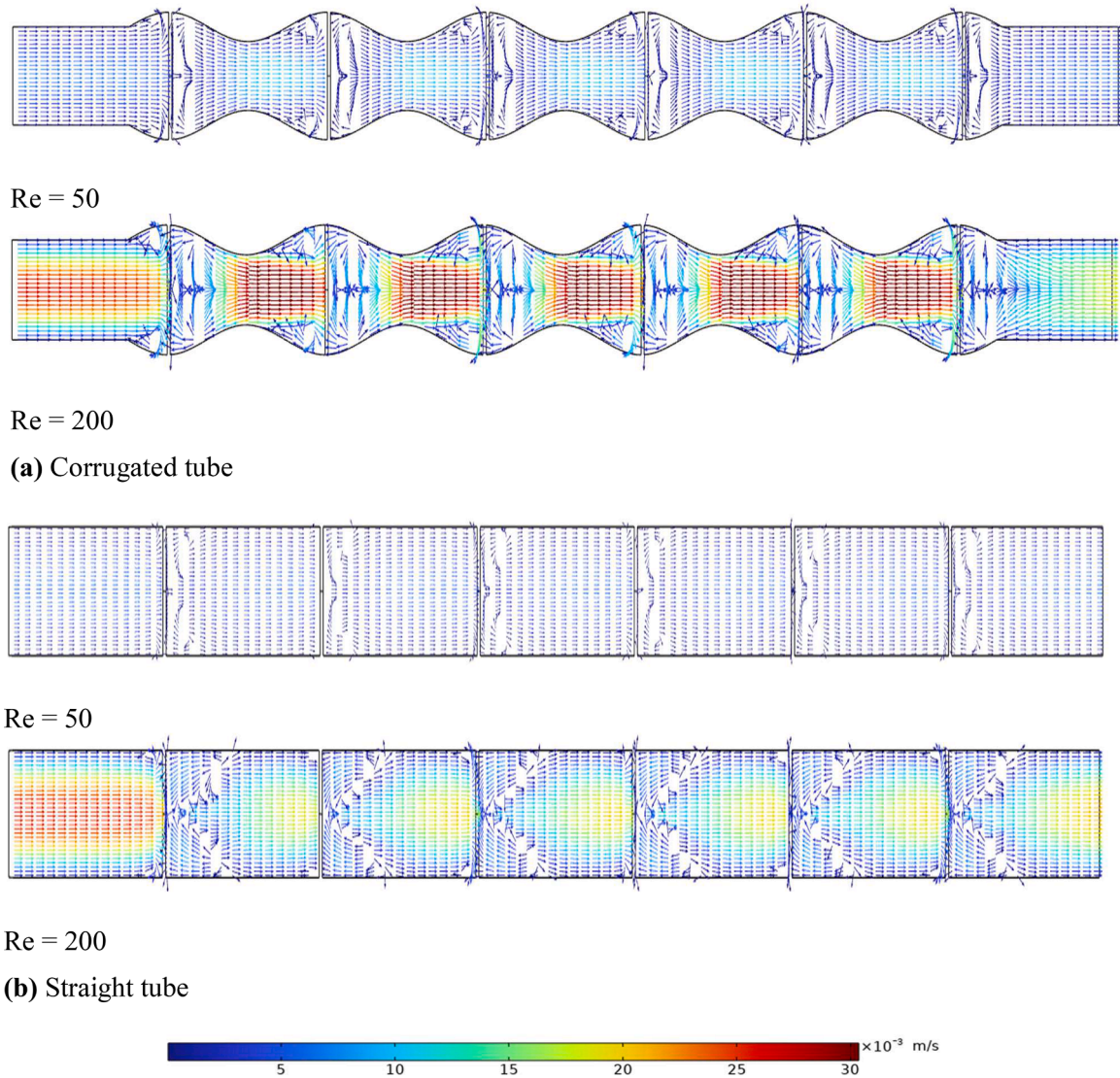


Fig. 8. Arrow plots of the velocity field at  $z/D = 0.5$  for 2-TA,  $\beta = 0^\circ$ .

straight tubes, especially when baffles are not used, due to hydraulic resistance [32]. The relationship between pressure drop and  $Re$  is illustrated in Fig. 7(b). It is observed that as  $Re$  number increases, pressure drop increases in both cases due to the increase in fluid velocity and flow vortex. The pressure drop with a corrugated tube is higher than that with a straight tube because the area between elements in the corrugated tube experiences greater wall shear caused by the throttling effect created by the converging zone. As depicted in Fig. 8, unlike a straight tube, which has a uniform tube diameter that produces a more uniform velocity profile, a corrugated tube has variations in tube diameter, leading to a less uniform velocity profile.

### 3.3. Influence of baffles

To investigate the impact of baffles on flow mixing in the proposed design at various  $Re$  values, a comparison was made between the mixing index of a static mixer with 2-TA and that of a static mixer with 4-TA. Fig. 9(a) illustrates the results, which indicate that the static mixer with 4-TA has a higher mixing index than the mixer with 2-TA in all  $Re$  cases, except for the highest value where they are equal. The augmenting  $Re$  number corresponds to an escalated mixing index, attributed to the expansion of the vortex pattern engendered behind the baffles. In the context of the 4-triangle arrangement (4-TA) configuration, four longitudinal vortices are manifested, in contrast to the 2-triangle arrangement (2-TA), as illustrated in Fig. 10, thereby yielding improved mixing performance. The findings demonstrate an elevation in the mixing index of 61.8%, 29.2%, and 25.1% at  $Re$  values of 50, 100, and 200, respectively. However, the mixing indices for all cases converge at low Reynolds numbers, implying a negligible influence of vortices and subdued chaotic flow dynamics.

Fig. 9(b) portrays the impact of baffles on the pressure drop within the mixer across diverse  $Re$  numbers. It is evident that the proposed design with 4-TA has a higher-pressure drop compared to the mixer with 2-TA at all  $Re$  cases, with increasing pressure drop values as  $Re$  number increases. The results indicate that the proposed design with 4-TA experiences a six-fold pressure drop increase compared to the mixer with 2-TA, mainly due to the throttling caused by the close distance between the triangles of the baffle.

### 3.4. Impact of staggered and aligned baffles

The effect of staggered and aligned baffles on the mixing index and pressure drop for 4-T and 2-T is illustrated in Figs. (11 and 12), respectively, across various  $Re$  numbers. The results indicate that an

improvement in  $Re$  value leads to an enhancement in the mixing index, however, the variation in mixing index for both cases remains almost the same for all Reynolds numbers. This is due to the converging zone of the corrugated pipe, which affects the efficacy of both staggered and aligned baffles by directing the flow towards the center of the element, as shown in Fig. 13. In addition to this, the element is occupying a zone large. Therefore, the velocity trajectory does not significantly change even when the element is rotated  $45^\circ$  in the case of staggered baffles. The pressure drop values display a similar trend in both configurations. Conversely, when comparing the 2-TA arrangement, distinctions in the mixing index values become evident. Particularly, the Mixing Index (MI) demonstrates a consistently ascending trajectory with rising  $Re$  number, yet the staggered configuration presents a higher mixing index in contrast to the aligned arrangement. This can be attributed to the confined area occupied by the element. Consequently, when the element undergoes a  $90^\circ$  rotation in the staggered configuration, flow redirection occurs, leading to amplified disturbances. Consequently, the mixing process becomes more dynamic at the expense of increased energy consumption.

It is frequently observed that using staggered baffles generates more turbulence, potentially resulting in a greater pressure drop compared to aligned baffles [33]. Nonetheless, several factors, including baffle spacing and placement, baffle shape, and fluid properties, impact the pressure drop in the proposed setup. To investigate the effect of the corrugated configuration in greater detail, Fig. 13 displays a cross-sectional view of the velocity vector field before the baffles. It can be observed that the convergence zone significantly contributes to driving particles into the center of the element, while the divergence zone creates a reversal flow, further emphasizing the impact of the corrugated configuration on flow behavior.

For an improved comprehension and analysis of the mixing process, a distribution of concentration streamlines has been supplied for diverse configurations, as depicted in (refer to Fig. 14). The  $Re$  number linked to the results illustrated in this figure is  $Re = 100$ . The mixing process of miscible fluids is significantly enhanced through the influence of chaotic advection, achieved by the incorporation of baffles and the generation of backflow within fluid streams. To enhance clarity, the concentration distributions are showcased across distinct cross-sectional planes.

Staggered baffles assume a pivotal role in promoting augmented backflow within fluid streams, a feature particularly pronounced in the 2-T configuration. This phenomenon engenders a larger interfacial area, thereby expediting the mixing process, as depicted in the concentration slices. Conversely, within the 4-T configuration, the introduction of a higher number of baffles fosters improved mixing in contrast to the 2-T

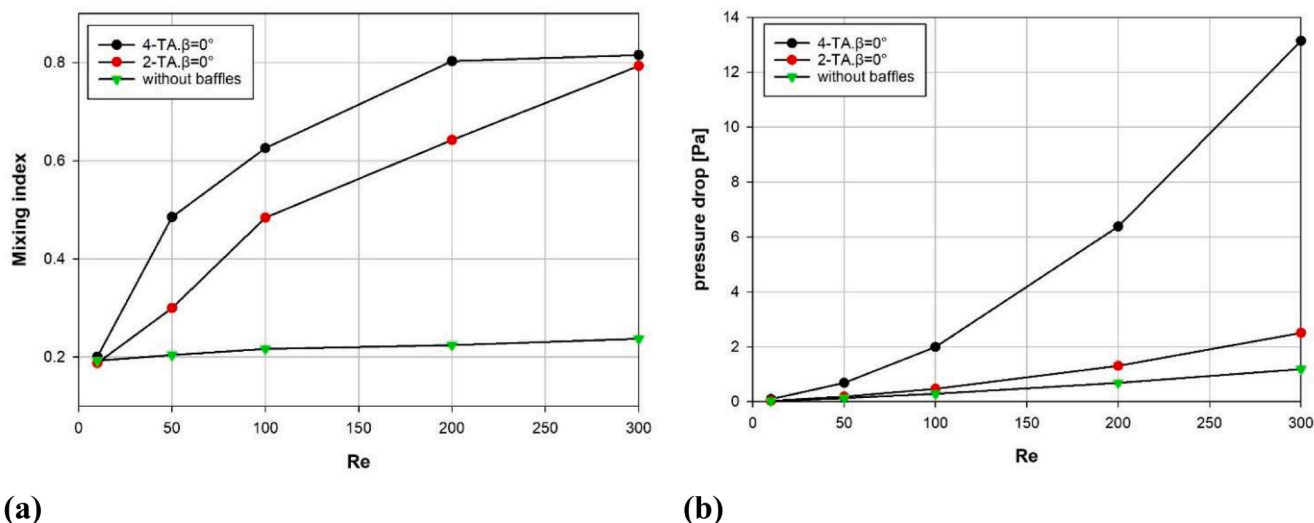


Fig. 9. Influence of number of baffles on (a) mixing index and (b) pressure drop of the static mixer with 4-TA,  $\beta = 0^\circ$ , 2-TA,  $\beta = 0^\circ$ , and without baffles.



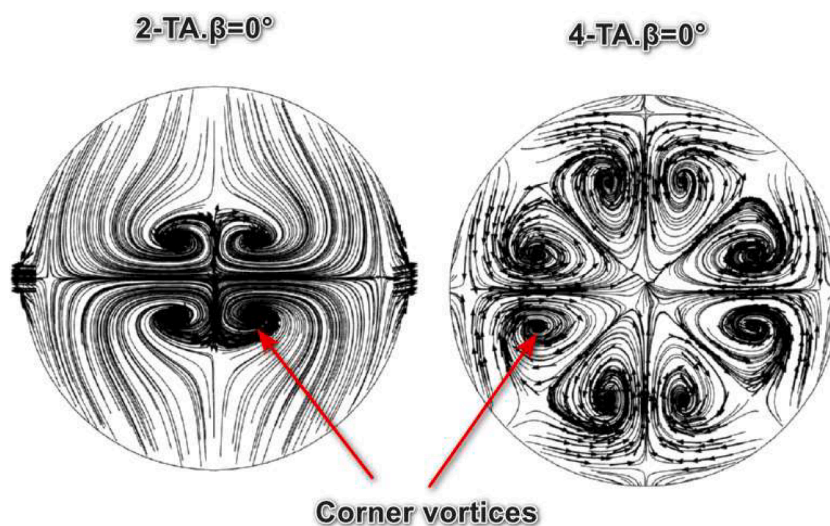


Fig. 10. Vertical cross-sectional streamlines behind the first element at  $Re = 100$ , showing the impact of the number of baffles on flow mixing.

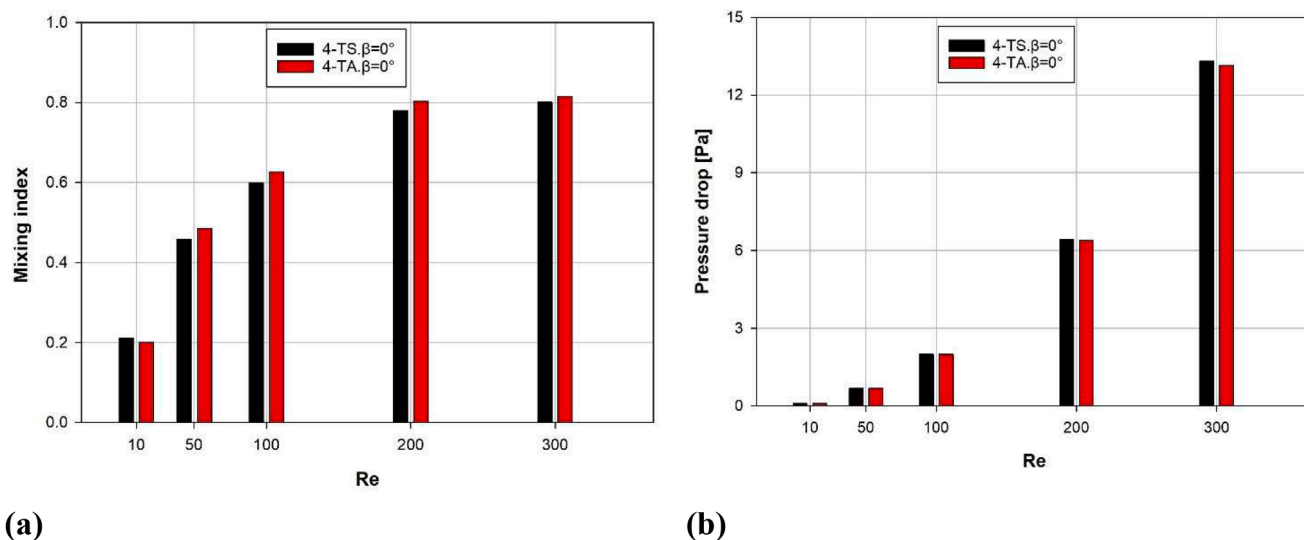


Fig. 11. Influence of staggered and aligned baffles on (a) mixing Index and (b) pressure drop for 4-triangular elements.

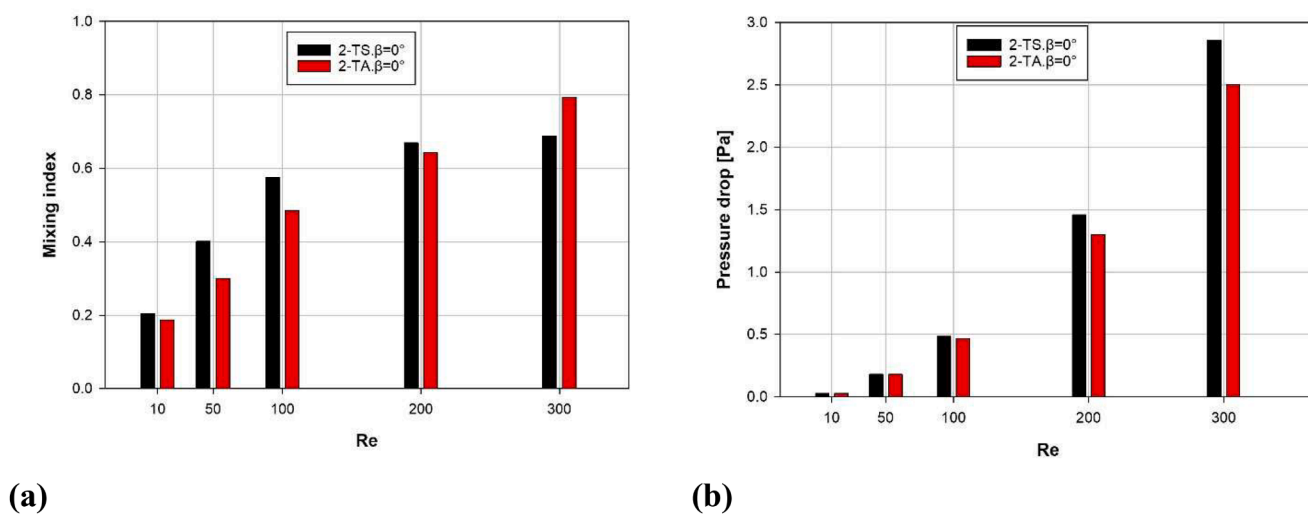


Fig. 12. Influence of staggered and aligned baffles on (a) mixing Index and (b) pressure drop for 2-triangular elements.

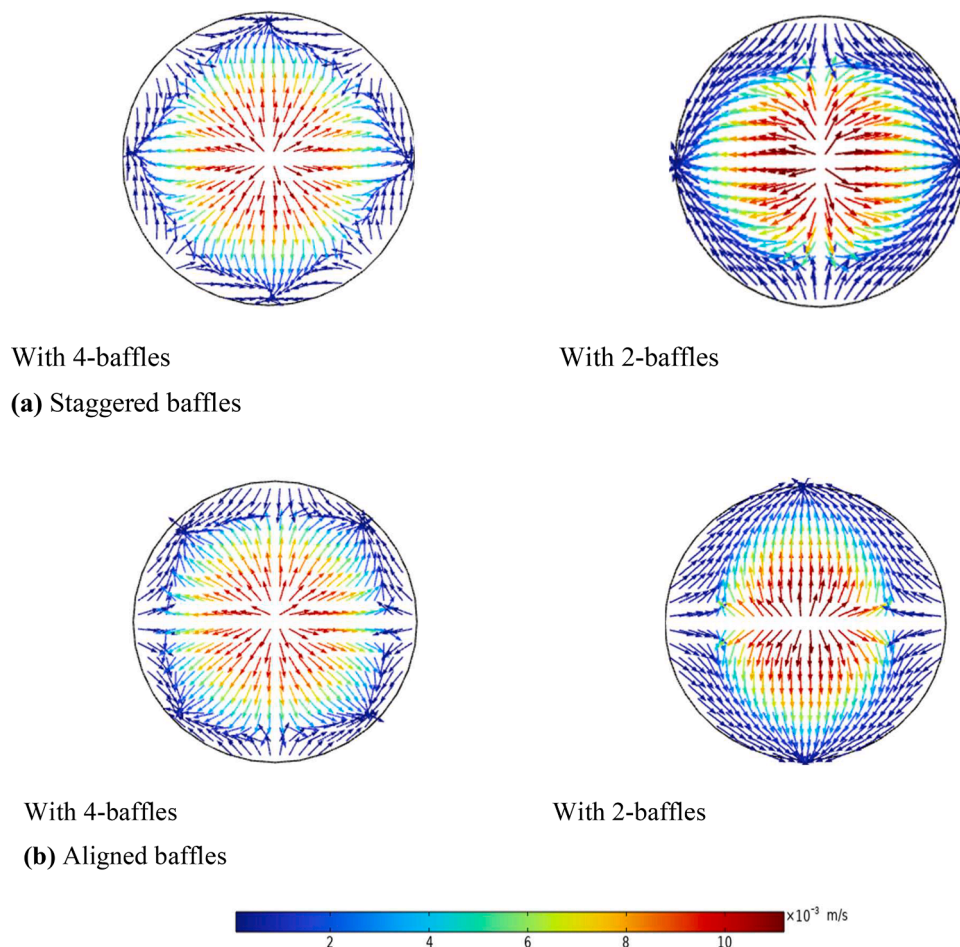


Fig. 13. Cross-sectional view of velocity vector field before the second element for Reynolds number ( $Re$ ) = 300.

arrangement. Nevertheless, the disparity in the mixing index remains fairly constant for both staggered and aligned configurations along the entire length of the mixer.

### 3.5. Impact of angle of attack

The addition of baffles to a static mixer system leads to the disturbance of flow patterns, the creation of vortices, and the effective mixing of fluid streams by increasing the chaotic area [14,34]. Nevertheless, this approach poses a challenge due to the considerable rise in pressure drop and energy consumption. This segment investigates and debates the influence of employing an angle of attack on the pressure drop and mixing efficacy.

A modification in the attack angle can influence both the mixing index and pressure drop. The effect of angle of attack on the mixing index is presented in Fig. 15(a), where the mixing index shows different behavior with varying Reynolds numbers for different angle of attack values. The minimum values of the mixing index were observed at  $\beta = 0^\circ$ , indicating that the fluid mixing was not optimal. However, when the angle of attack was increased to  $15^\circ$  and  $30^\circ$ , the mixing index increased significantly, reaching a maximum value of 0.99, which represents a 22.6% improvement compared to the  $\beta = 0^\circ$  case. This improvement is attributed to the swirling motion caused by the rotation of fluid particles around the central axis, which enhances the mixing efficiency.

The maximum tangential velocities are located near the channel wall, as depicted in Fig. 16, which displays the distribution of tangential velocities in the channel's cross-section. On the other hand, when the angle of attack was increased to  $45^\circ$ , the flow rotation was less drastic, resulting in decreased mixing index values, particularly at low

Renumbers.

Fig. 15(b) illustrates the impact of angle of attack on pressure drop. The results reveal that each angle of attack has a distinct pressure drop value and follows the same trend within the examined  $Re$  range. As depicted in the figure, there is an observable trend wherein the pressure drop diminishes proportionally with an increment in the angle of attack. Notably, the maximal pressure magnitude is documented at  $\beta = 0^\circ$ . This phenomenon can be rationalized by the premise that a reduced angle of attack frequently precipitates heightened fluid collision and turbulence, thereby augmenting flow impedance and contributing to an elevated pressure drop. Conversely, with an elevation in the angle of attack, fluid particles acquire a heightened propensity for circumferential rotation about the central axis, engendering a vortical motion that serves to amplify flow efficacy and diminish pressure drop.

### 3.6. Optimization

In this investigation, the impact of several factors, including  $Re$  number, corrugated static mixer configuration, baffle count and positioning, and baffle angle of attack, was assessed to determine their influence on the pressure drop and mixing quality. The objective was to identify the most favorable geometrical setup for the system based on the optimization findings from all the cases analyzed.

The findings of the study showed that increasing the Reynolds number had a positive impact on mixing quality. Fig. 15 illustrates that, for a case of  $\beta = 0^\circ$ , the mixing index reaches a steady value when the Reynolds number exceeds 100. This phenomenon can be attributed to the shape and angle of the baffle, which results in a relatively low-turbulence flow when the fluid flows perpendicularly on the baffles.

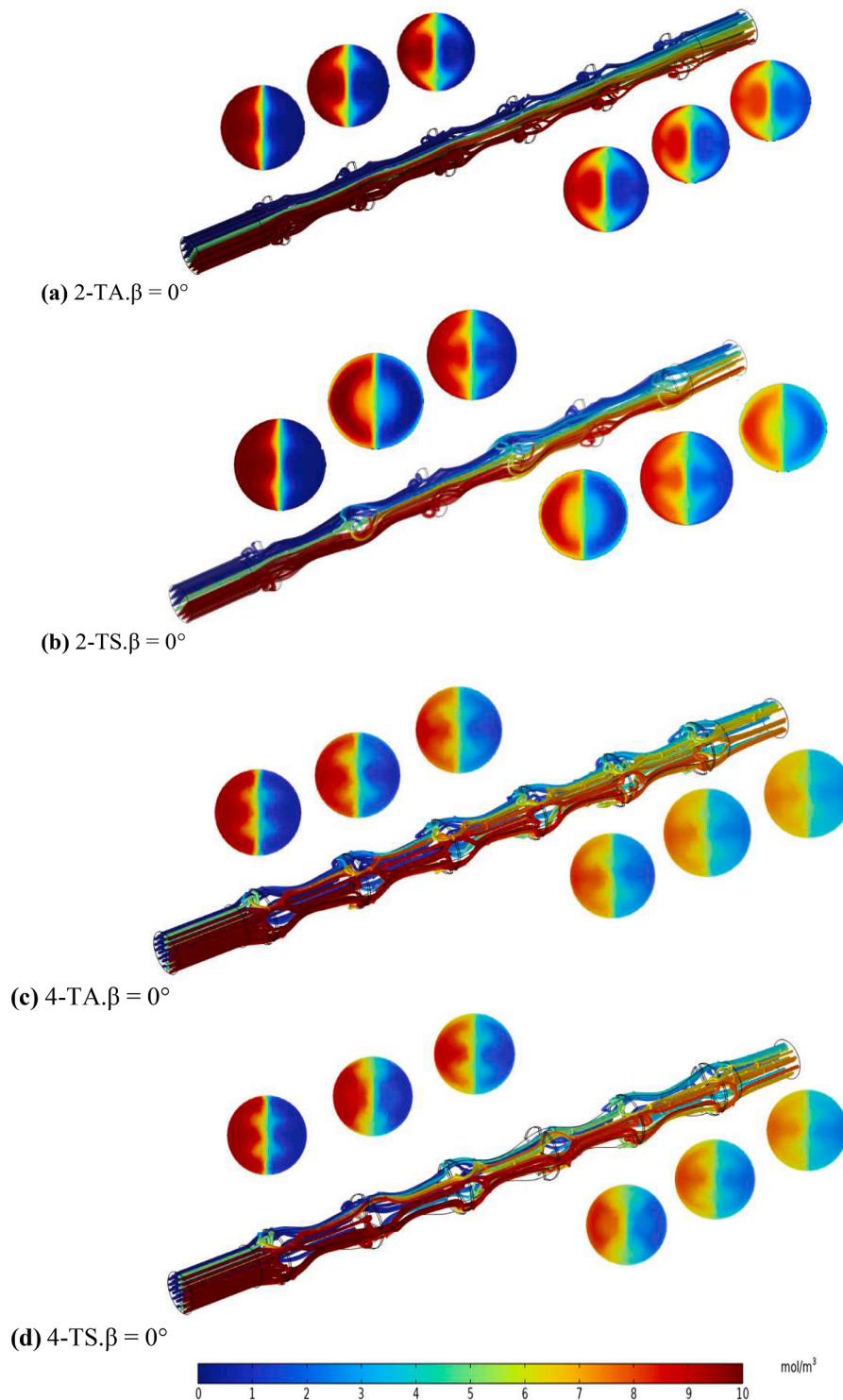


Fig. 14. Streamline distribution of concentration along static mixers for  $Re = 100$ .

However, changing the angle of attack to  $15^\circ$  or  $30^\circ$  leads to a considerable increase in mixing efficiency, even for low Reynolds numbers where turbulence is not typically high enough to promote efficient mixing. This is likely due to the increased creation of swirls in the fluid, generating a central vortex that contributes to mixing the fluid near the channel wall. The maximum tangential vortices in these areas enhance the mixing process by swirl flow and increasing the interfacial area.

To provide a more detailed characterization of the mixing process, Fig. 17 presents concentration distributions at various cross sections behind static mixer elements with different angles of attack for  $Re =$

100. As depicted in the figure, the first and second mixer elements have a substantial impact on the mixing process, resulting in improved mixing following the second unit, particularly when  $\beta$  is set to  $15^\circ$  or  $30^\circ$ , compared to other configurations. This enhancement in mixing is due to the creation of more robust tangential vortices and swirling motion in the fluid, resulting from the angle of attack, which stimulates the mixing process. Nonetheless, elevating the angle of attack to  $45^\circ$  does not increase the mixing efficiency as much as lower angles of attack, such as  $15^\circ$  or  $30^\circ$ . Our observations suggest that, in this case, the fluid becomes homogenized only after the final element.

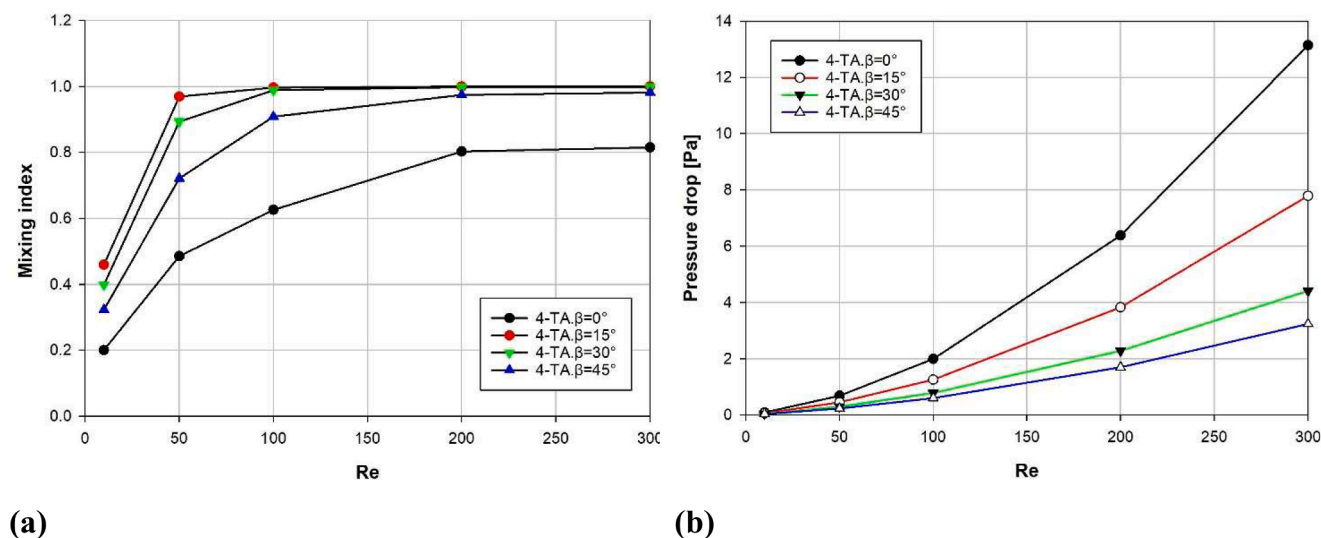


Fig. 15. Influence of the angle of attack on (a) mixing Index and (b) pressure drop for 4-triangular elements.

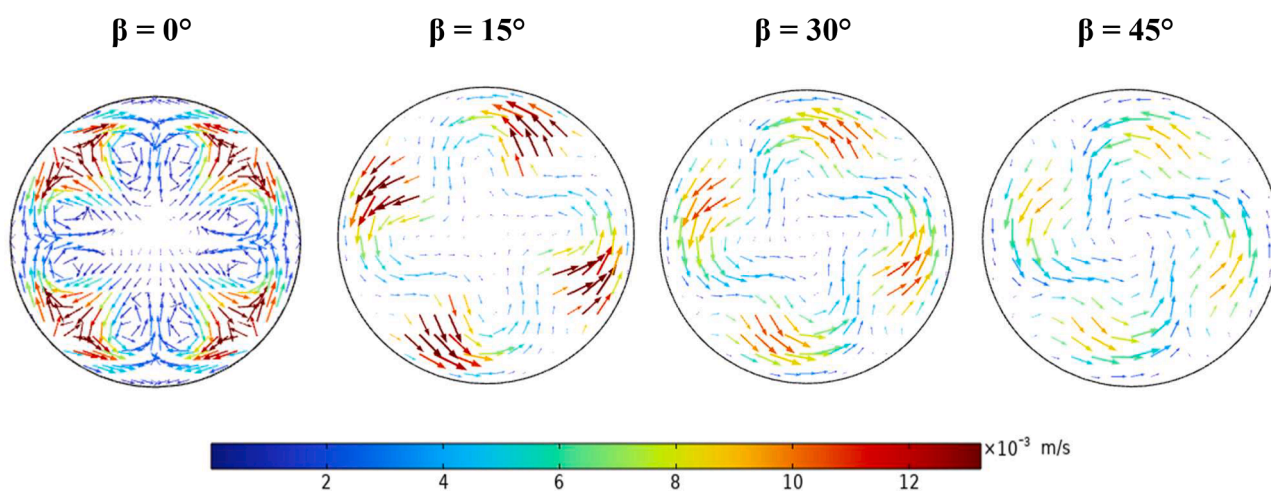


Fig. 16. Velocity field in the cross-sectional plane behind the second element at  $Re = 100$ .

To assess the effectiveness of the static mixer configurations, two parameters were used: mixing index and pressure drop. Fig. 18 delineates the influence of the angle of attack on these parameters at a  $Re$  number of 100. A progressive enhancement in the mixing index value was observed with the escalation of the angle of attack, evident for both  $\beta = 15^\circ$  and  $\beta = 30^\circ$ , culminating in a peak value of 0.99. This pinnacle value signifies a notable 22.6% amelioration when juxtaposed with the initial case. Nonetheless, surpassing an angle of  $30^\circ$  engendered a decline in the mixing index. Concurrently, an ascent in the angle of attack correlated with a reduction in the pressure drop magnitude. A comparative assessment of the mixing index and pressure drop outcomes deduced that the optimal angle of attack for the given static mixer configuration is  $\beta = 30^\circ$ . This angle elicits the highest mixing index while concurrently minimizing energy consumption.

#### 4. Conclusions

This research utilized numerical simulations to examine the mixing behavior in corrugated tubes featuring triangular baffles within Newtonian flows. The study aimed to investigate the effects of the angle of attack, number of baffles, and their placement on the mixing index improvement and pressure drop reduction over a wide range of  $Re$

numbers from 10 to 300. Based on the simulation results, the study drew the following key findings:

- The study investigated various configurations and found that higher  $Re$  numbers resulted in increased mixing index due to the generation of vortices and chaotic flow behavior. The results indicated that higher turbulence intensity resulting from increased  $Re$  number contributed to a more chaotic flow and improved mixing efficiency. Consequently, it can be inferred that enhancing the  $Re$  number plays a crucial role in improving the mixing performance of triangular baffle-equipped corrugated tubes.
- The numerical analysis clearly demonstrated that the use of corrugated tubes with triangular baffles had a significant impact on the mixing process, as evidenced by a remarkable three times increase in the mixing index compared to a straight tube. This substantial improvement can be attributed to the presence of the triangular baffles, which generated a series of eddies and vortices that induced greater turbulence in the flow, thus increasing the interfacial area between the fluids and enhancing mixing.
- When comparing the 4-TA and 2-TA configurations, it was found that the number of baffles had a noticeable influence on the mixing index, with the 4-TA configuration demonstrating a great increase in the



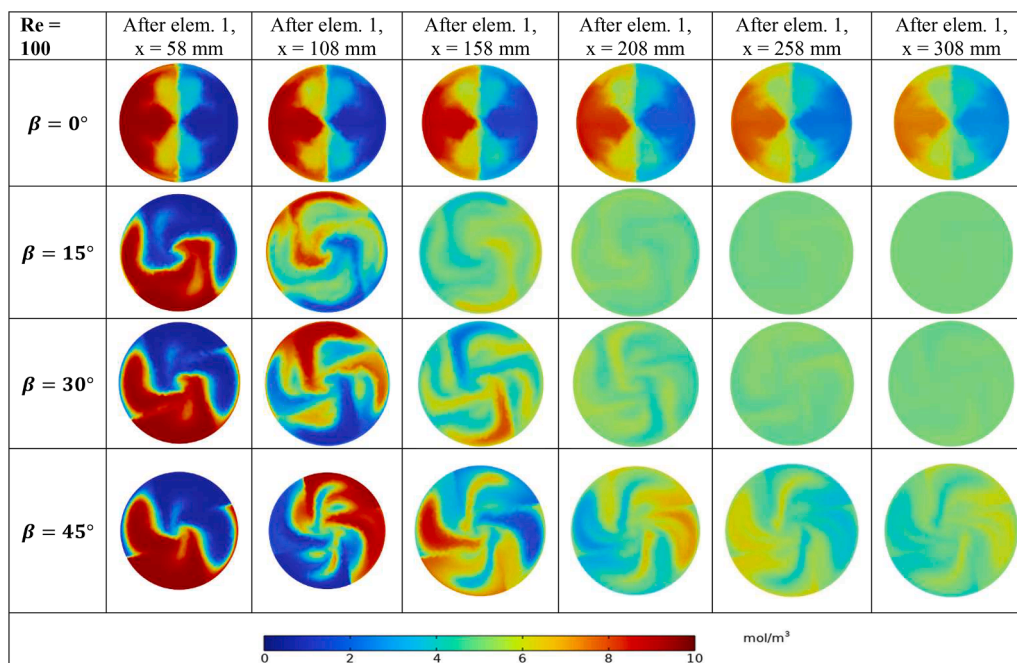


Fig. 17. The concentration distributions at different cross sections behind static mixer elements with various angles of attack for Reynolds number ( $Re = 100$ ).

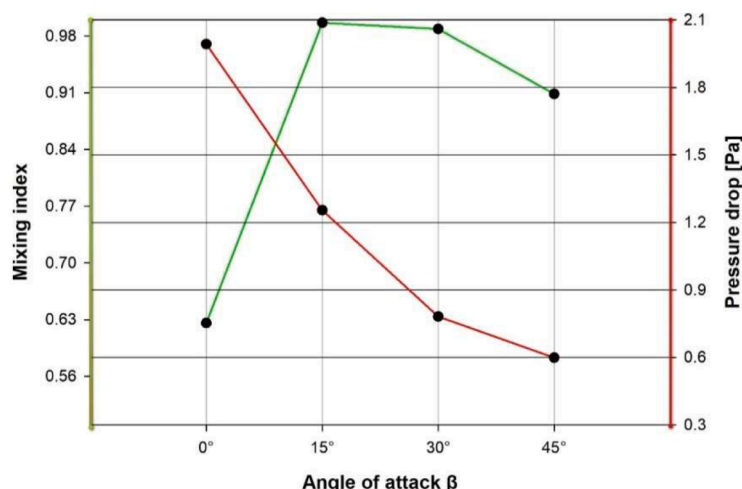


Fig. 18. Influence of angle of attack on mixing index and pressure drop for Reynolds number of 100.

mixing index compared to the 2-TA. However, this came at the expense of higher energy consumption and an increase in pressure drop.

- The research findings suggest that the staggered and aligned baffle configurations for the 4-T case have similar mixing efficiency and power consumption. However, the angle of attack was found to have a significant effect on the flow pattern and increase the mixing rate by 22.6% when compared to the initial case. Consequently, the study concluded that the optimal configuration was achieved with an angle of  $30^\circ$ .
- By adjusting the angle of attack to  $15^\circ$  and  $30^\circ$ , the mixing process significantly improved, resulting in complete homogeneity right after the second element.

In conclusion, the outcomes of this investigation underscore the potential utilization of a corrugated tube fitted with triangular baffles as a viable approach for enhancing the hydrodynamic efficiency of static mixers, concomitantly leading to a reduction in energy consumption.

Here are some potential directions for future research:

- Investigate the influence of different fluid properties, such as viscosity and density, on the performance of the static mixers.
- Implement optimization algorithms to systematically explore the design space of corrugated tubes, triangular baffles, and vortex generators.
- Extend the study to explore the mixing performance of the designed static mixer configurations in transitional and turbulent flow regimes.
- Extend the investigation to include scenarios where multiple components or phases are being mixed.
- Study the scalability of the proposed static mixer configurations to larger industrial scales.
- Explore advanced numerical methods, such as lattice Boltzmann simulations or other high-fidelity techniques, to provide more accurate predictions of the fluid dynamics and mixing behavior within the static mixers.

- Conduct a comprehensive sensitivity analysis to identify the most influential design parameters and fluid properties on mixing efficiency.
- Investigate the potential synergies of combining the proposed static mixer configurations with other mixing techniques, such as active mixing methods (e.g., ultrasonic agitation) or additional passive elements (e.g., helical inserts), to further enhance mixing efficiency.
- Assess the environmental implications of using the optimized static mixer configurations, considering factors like material choice, energy consumption, and waste production.

Each of these future research directions has the potential to expand the knowledge and practical applications of improving mixing efficiency in laminar-flow static mixers with baffle inserts and vortex generators using corrugated tubes.

## Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

## Informed consent

Informed consent was obtained from all individual participants included in the study.

## CRediT authorship contribution statement

**Ezzine Bennour:** Conceptualization, Methodology, Writing – review & editing. **Cheikh Kezrane:** Data curation, Writing – original draft, Visualization. **Noureddine Kaid:** Investigation, Writing – review & editing. **Sultan Alqahtani:** Data curation, Formal analysis. **Sultan Alshehry:** Formal analysis, Investigation, Resources. **Younes Menni:** Resources, Validation, Data curation.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

No data was used for the research described in the article.

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